THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** normal	Girls often grow earlier.
2	**(a)** copper	Best conductor.
3	**(c)** 60 km/h	90/1.5.
4	**(b)** mouth, oesophagus, stomach, small intestine, large intestine	Standard order.
5	**(d)** growth spurt	Both grow taller.
6	**(c)** lower than over sea	Warm air rises, low pressure.
7	**(c)** 50 km/h	100/2.
8	**(c)** 960	16 x 60.
9	**(d)** neck	Front of the neck.
10	**(d)** white loose cotton	Reflect and breathe.
11	**(b)** speed	Rise over run.
12	**(b)** four chambers	Mammal.
13	**(b)** larynx is growing	Voice box.
14	**(b)** radiation	Air is a poor conductor.
15	**(d)** 9 km/h	1.5 x 6.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Larynx grows more in boys (1); Testosterone causes this (1).	1+1
17	Metals are good conductors of heat (1); Wood is an insulator (1).	1+1
18	150/2.5 = 60 km/h (1); 60 x 5/18 = 16.7 m/s (approx.) (1).	1+1
19	Muscles need more energy and oxygen (1); More carbon dioxide has to be removed (1).	1+1
20	Strong muscles and bones (1); Better mood (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Obesity risk (1); Low vitamins and minerals (1); Add fruits, vegetables, milk (1).	1+1+1
22	Radiation to nearby objects (1); Convection moves warm air (1); Conduction in the heating element (1).	1+1+1
23	40 min = 2/3 h (1); 12 / (2/3) = 18 km/h (1); 5 m/s (1).	1+1+1
24	Mouth: starch begins digestion (1); Stomach and small intestine: digestion completed (1); Large intestine: water removed, undigested egested (1).	1+1+1
	OR: Peer pressure (1); Curiosity (1); Awareness and refusal skills (1).	1+1+1
25	Metal conducts (1); Plastic insulates (1); Wool reduces heat loss (1).	1+1+1
	OR: Going 3/5 = 0.6 h (1); Returning 3/3 = 1 h (1); Total 1.6 h (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Boys: testosterone (1); Voice, beard (1); Girls: oestrogen (1); Breast, periods (1); Both: growth spurt (1).	1+1+1+1+1
	OR: Conduction: wax pins on rod (1); Convection: permanganate crystals in water (1); Radiation: hand near a lamp (1); Observations noted (1); Conclusions drawn (1).	1+1+1+1+1
27	Total time 4 h (1); Average 240/4 = 60 km/h (1); Running time 3.5 h (1); 240/3.5 = 68.6 km/h (1); Running speed is higher (1).	1+1+1+1+1

	OR: Mouth: amylase on starch (1); Stomach: pepsin on proteins (1); Small intestine: bile for fat, pancreatic juice for all (1); End products: sugar, amino acids, fatty acids (1); Absorbed by villi (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** starch is delayed	Amylase missing.
2	**(b)** fast growth needs more energy	Energy.
3	**(a)** sea	Sea is warmer at night.
4	**(b)** 2 h	60/30.
5	**(a)** increases	More oxygen needed.
6	**(a)** abdomen	Near the stomach.
7	**(a)** layers of woollen clothes	Trapped air.
8	**(b)** 5 m/s	20/4.
9	**(a)** arteries	Goes to lungs.
10	**(c)** goitre	Thyroid.
11	**(d)** convection	Hot air rises.
12	**(c)** 10 m/s	36 x 5/18.
13	**(a)** breaking fat into tiny droplets	Emulsification.
14	**(a)** fruit and nuts	Healthy.
15	**(c)** water takes longer to heat	High heat capacity.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Copper is a good conductor (1); Even and fast heating (1).	1+1
17	Effort and gravity act against uphill (1); Speed changes so motion is non-uniform (1).	1+1
18	Arteries have thick elastic walls (1); Veins have thin walls and valves (1).	1+1
19	Part of the body's cycle (1); Prepares for possible pregnancy (1).	1+1
20	Heat flows from the fish to the ice (1); The fish stays cold (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Damages lungs (1); Causes cough and cancer (1); Reduces stamina (1).	1+1+1
22	Oil glands active (1); Dead cells block pores (1); Bacteria (1).	1+1+1
23	Hot coal radiates (1); Food absorbs (1); Convection also helps (1).	1+1+1
24	First 0.5 m/s (1); Second 0.75 m/s (1); Speed increased (1).	1+1+1
	OR: Constipation (1); Acidity (1); Indigestion (1).	1+1+1
25	Egg matures (1); Lining thickens (1); If no fertilisation, lining sheds (1).	1+1+1
	OR: Land is hottest then (1); Larger temperature difference (1); Strong convection (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Mood swings (1); Need for acceptance (1); Talk to family (1); Hobbies (1); Seek help if needed (1).	1+1+1+1+1
	OR: Black pipes absorb (1); Glass cover traps heat (1); Water circulates by convection (1); Hot water tank insulated (1); Saves fuel (1).	1+1+1+1+1
27	Friction and air resistance (1); Amplitude decreases (1); Spring or weights give energy (1); Regular winding (1); Clock continues (1).	1+1+1+1+1
	OR: Humans: lungs (1); Fish: gills (1); Insects: tracheae (1); Earthworm: moist skin (1); Amoeba: surface of the body (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 8 m/s	100/12.5.
2	**(d)** decreases	Air goes out.
3	**(b)** kidneys	Small glands.
4	**(b)** air gets trapped between layers	Insulation.
5	**(c)** faster	10 m/s versus 5 m/s.
6	**(d)** veins	Veins.
7	**(d)** diabetes	Sugar high.
8	**(a)** radiation	Sideways transfer.
9	**(b)** 90 km/h	25 x 18/5.
10	**(d)** pepsin: proteins	Pepsin acts on proteins.
11	**(c)** loss of milk teeth at 5	Teeth shedding starts earlier.
12	**(a)** water cools slowly	Takes more time to cool.
13	**(a)** half	Time inversely proportional.
14	**(a)** oxygen	Oxygen is used.
15	**(a)** brain	Brain.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Larger angles can change motion (1); Time period measurement is easier (1).	1+1
17	More water is lost as sweat (1); Kidneys conserve water (1).	1+1
18	It releases hormones that stimulate testes or ovaries (1); It controls growth (1).	1+1
19	Hot air inside is lighter than the surrounding air (1); Convection lifts it (1).	1+1
20	Time = 225/90 (1); 2.5 h (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	200/20 = 10 m/s (1); 10 x 18/5 (1); 36 km/h (1).	1+1+1
22	Blow through a straw into lime water (1); It turns milky (1); CO2 is present (1).	1+1+1
23	Endocrine: no ducts (1); Digestive: ducts carry juices (1); Examples: thyroid vs salivary (1).	1+1+1
24	Heat conducts to both ends (1); Equal for equal distance (1); Wax pins fall from the centre outward (1).	1+1+1
	OR: Traffic (1); Signals (1); Stops and starts (1).	1+1+1
25	Spiracles open to the outside (1); Tracheae carry air (1); Direct to cells (1).	1+1+1
	OR: Guidance (1); Reduces stress (1); Prevents wrong steps (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Nutrition for growth (1); Sleep for rest (1); Hygiene for health (1); Exercise for fitness (1); All support mental health (1).	1+1+1+1+1
	OR: Vacuum prevents conduction and convection (1); Silvered walls prevent radiation (1); Loss of vacuum lets air carry heat (1); Contents lose heat faster (1); Flask useless (1).	1+1+1+1+1
27	40 km in first 2 h (1); Horizontal for 1 h (1); 60 km more in 2 h (1); Total 100 km in 5 h (1); Average 20 km/h (1).	1+1+1+1+1
	OR: Four chambers (1); Right side deoxygenated (1); Lungs oxygenate (1); Left side pumps to body (1); Blood returns (1).	1+1+1+1+1

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